



California State University, Northridge Project Report

Adhunika kandala
Madhupriya Allenki
Long Nguyen
Siri Palnati

Under the Guidance of

Professor. Akash Gupta
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Data Mining and Predictive Analytics for Business

A comprehensive report & python code on Nursing Homes

Executive Summary:

This project analyzes what drives gross revenue in U.S. nursing homes from 2015 to 2021 using linear and random forest regression models. We used data on finances, quality, penalties, and ownership. Random Forest performed better ($R^2 = 0.6664$), identifying Number of Beds, Occupancy Rate, and Certified Beds as key factors. Linear regression ($R^2 = 0.558$) showed similar results. We found that operational factors like facility size and patient volume are more important than ownership type or penalties. Revenue grew steadily until 2019 but became unstable during COVID-19. To stay financially strong, nursing homes should focus on keeping beds full, managing debt, and using capacity efficiently.

Introduction:

The long-term care industry, particularly nursing homes, plays a vital role in supporting the healthcare needs of elderly and chronically ill individuals. With the continued growth of the aging population in the United States, demand for quality and sustainable nursing home care has risen substantially. The changing demographics, coupled with evolving healthcare policies and unforeseen crises such as the COVID-19 pandemic, have heightened the significance of an overall examination of the nursing home industry.

This project aims to assess the overall financial performance of U.S. nursing homes from 2015 to 2021 and identify the key drivers behind that performance. In addition to assessing revenue, expenses, and profitability trends, the project also investigates how facility size, location, and ownership type influence financial outcomes. The impact of COVID-19, particularly in 2020 and 2021, is also analyzed to determine its short-term and long-term effects on the industry.

Beyond financial metrics, the project considers operational influences such as quality of care, reported health deficiencies, and penalties — all of which shape the sustainability and public trust in nursing home operations. By integrating these various dimensions, the analysis provides a data-driven foundation for making informed recommendations regarding potential investment opportunities, operational improvements, and policy considerations.

Objectives:

1. Evaluate the overall financial performance of nursing homes in the United States during this period.
2. Identify the influential factors affecting the financial performance of nursing homes.
3. Determine the most significant factors impacting the performance of nursing homes.
4. Describe the trends in performance and these influential factors.
5. Analyze the impact of COVID-19.

Data Description:

This project uses several publicly released U.S. nursing home datasets covering 2015 to 2021. The datasets contain financial reports, facility information, quality measures, health deficiencies, penalties, and COVID-19 vaccination data. They were primarily sourced from the Centers for Medicare & Medicaid Services (CMS) and later combined into one analytical dataset after cleaning, standardization, and merging.

- **Cost Reports**

The cost report data provides detailed financial information for each nursing home, including total patient revenues, total operating expenses, net profit, inpatient PPS amounts, total other revenues, and profit margins. These amounts were aggregated and analyzed by year to look at overall financial performance and profitability trends.

- **Provider Info**

This database includes structural and organizational characteristics of each facility such as ownership category (for-profit, nonprofit, or government), certified bed size, urban/rural designation, and geographic area. These variables were used to analyze how structural factors influence financial outcomes.

- **Health Deficiencies**

Health deficiency data records the number and severity of violations identified during regulatory inspections. These include deficiencies in patient care, staffing, infection control, and safety. The total number of deficiencies per facility was used as a measure of care quality risk.

- **Penalties**

The Penalties dataset contains information about civil monetary penalties issued to nursing homes due to violations of federal care standards. Each record includes the

CMS Certification Number (CCN) to identify the facility, the date of the penalty, the amount of the fine, and the type of enforcement action taken.

- **Quality Measures**

The quality dataset includes performance scores on patient outcomes, clinical measures, and care processes. These scores were used to evaluate whether higher quality of care is associated with better financial performance.

- **COVID-19 Vaccination Data**

The COVID-19 Vaccination dataset includes facility-level data on vaccination rates among nursing home residents and staff during the years 2020 and 2021. Key variables in this dataset include the percentage of residents vaccinated, percentage of staff vaccinated, and the reporting dates.

Methodology:

For this project, we used Python to clean, combine, and analyze data from U.S. nursing homes spanning 2015 to 2021. We used regression models to identify the key factors influencing the financial performance of nursing homes.

Tools

- Pandas and NumPy for data manipulation and preprocessing
- Scikit-learn and Statsmodels for building and evaluating regression models
- Matplotlib and Seaborn for visualizing trends, model performance, and feature importance

Data Cleaning

The data includes many useful attributes explaining various categories that requires concatenation and merge. We realized that Provider ID (SNF identification number) is available across all datasets, thus, aggregation is performed sequentially as needed in order to generate useful information. When analyzing specification of each dataset, the problem was that they include roughly 7.8% missing values on useful attributes which could significantly impact analytic statistics. On the other hand, Cost Report and Provider Info datasets have a relatively high number of outliers displayed in boxplot using IQR filtering and quantile threshold.

- **Missing Values:** We explored different imputation methods to handle missing data. Since the datasets had many outliers, we chose median imputation instead of the mean. Median values are less affected by extreme values and help improve correlation and model stability during analysis.

- **Outlier Handling:** We first used capping to limit extreme values by replacing outliers with the 0.5th and 99.5th percentile values. This helped preserve more data, especially when record counts were low. However, we noticed that removing outliers instead of capping them led to better model performance. Specifically, R^2 scores improved after removing highly skewed data points, so we adopted outlier removal for our final analysis.
- **Data Formatting:** ID column naming was standardized to maintain consistent formats, and categorical variables (i.e. ownership type) were transformed using label encoding for better performance in machine learning models.
- Although multiple datasets were available, we focused only on the Cost Report, Provider Info, Quality Measures, and Penalties datasets. These were the most relevant and reliable for understanding financial and operational performance. Other datasets were not used due to incomplete data or limited relevance to our analysis.

Creating New Features:

To enhance the model's predictive capabilities, we created several new features:

- Profit Margin = $(\text{Net Income} / \text{Gross Revenue}) \times 100$
- Occupancy Rate = $(\text{Total Days of Care} / (\text{Bed Count} \times 365)) \times 100$
- Total Fine Amount = Aggregated penalty amounts per facility per year
- Debt Ratio = $\text{Total Liabilities} / \text{Total Assets}$

Models we used: We built and tested two regression models to understand which factors affect nursing home gross revenue:

Multiple Linear Regression (MLR)

We first used MLR to analyze the linear relationships between selected features and gross revenue. This model achieved a R^2 score of **0.558**, meaning it explained approximately 55.8% of the variation in gross revenue. The model provided clear coefficients for each feature, allowing us to interpret their direction and magnitude of influence. Features such as BEDCERT, OWNERSHIP, and Occupancy Rate had strong positive effects, while variables like Debt Ratio showed negative impacts.

Random Forest Regression

To capture more complex, nonlinear relationships between variables and gross revenue, we applied a Random Forest Regressor. This model improved performance over MLR, achieving an R^2 score of 0.6664. Instead of coefficients, this model provides feature importance scores. It identified Number of Beds (31.9%), Occupancy Rate (25.7%), BEDCERT (13.8%), and RESTOT (11.2%) as the most influential predictors. Features like Quality Score and Ownership contributed minimally. Random Forest proved better

in handling complex interactions among variables, making it a strong candidate for predictive modeling in this context.

Given its better fit and ability to handle nonlinear relationships, we ultimately chose Random Forest as our final model.

Analysis and Findings

1. Evaluate the overall financial performance of nursing homes in the United States during this period.

To analyze financial trends in U.S. nursing homes from 2015 to 2021, we focused on five key variables from the Cost Report dataset.

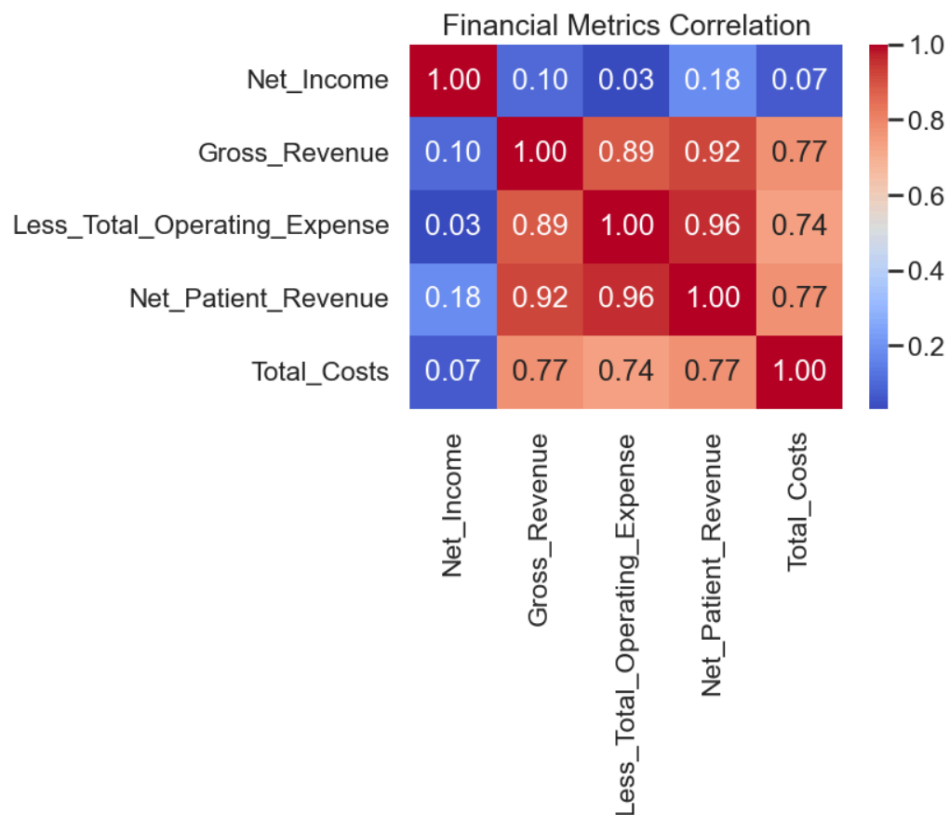
- **Gross Revenue:** Captures total earnings before deductions, giving a sense of each facility's scale and service volume.
- **Less Total Operating Expense:** Captures total earnings before deductions, giving an idea of each facility's scale and service volume.
- **Net Income:** The total profit or loss for the facility after deducting operating expenses from gross revenue.
- **Net Patient Revenue:** Isolates revenue directly tied to patient care, which is the primary operational driver for nursing homes. It reflects reimbursement levels and occupancy.
- **Total Costs:** Broadens the scope beyond operating expenses to include all cost centers (e.g., administrative, overhead), giving a fuller picture of expenditure patterns.

These five indicators together provide a comprehensive and focused view of each facility's revenue, cost efficiency, and financial understanding over time. Other available variables, such as fixed assets, equipment, admissions, and discharges are important operational metrics, but they are either more indirect or consistent for statistical financial analysis.

Correlation

The correlation analysis revealed that while revenue and patient-related income are closely linked to operating costs, net income shows only weak association with any core financial metric. This suggests that profitability is likely influenced by external or administrative factors (e.g., management efficiency, government reimbursements, or penalties) rather than just scale of operations. The near-perfect alignment between net patient revenue and operating expenses highlights how tightly nursing homes operate

on cost-recovery margins. These patterns also indicate that many nursing homes scale revenue through patient volume.



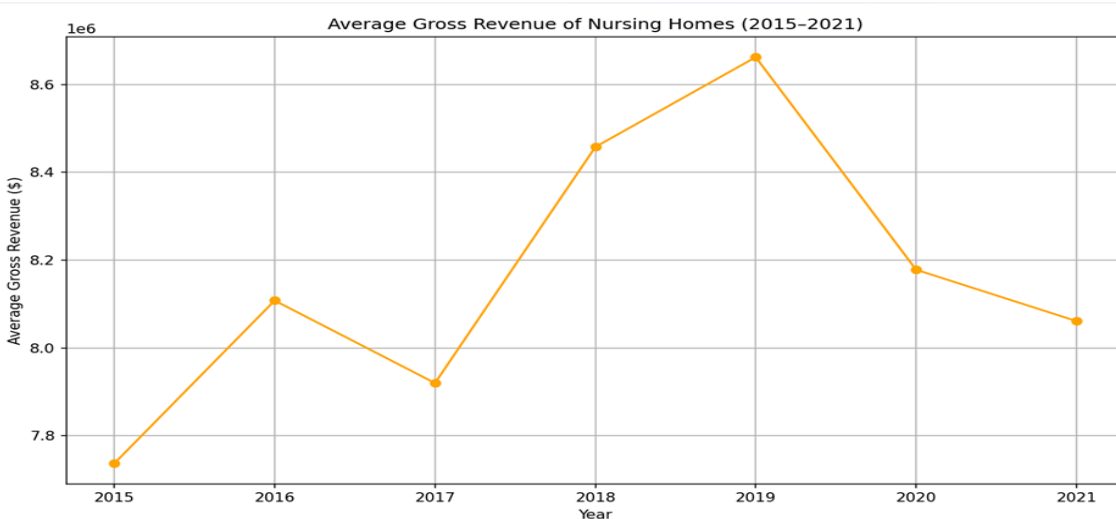
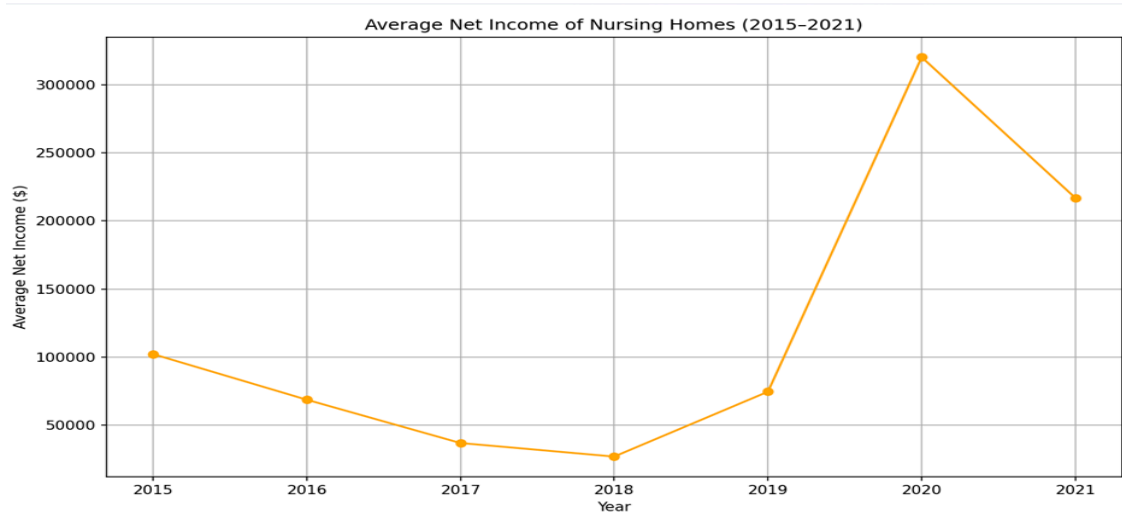
The correlation heatmap above highlights strong linear relationships between several financial metrics. Notably, *Gross_Revenue*, *Net_Patient_Revenue*, and *Less_Total_Operating_Expense* exhibit extremely high correlation coefficients exceeding 0.89 with one another. This suggests they carry redundant information and could introduce multicollinearity issues in predictive models. To reduce redundancy and improve model efficiency and interpretability, we selected only one representative variable *Gross_Revenue* to capture the shared variance among these highly correlated features. By doing so, we preserve essential financial information while minimizing overfitting risks and maintaining a cleaner feature set.

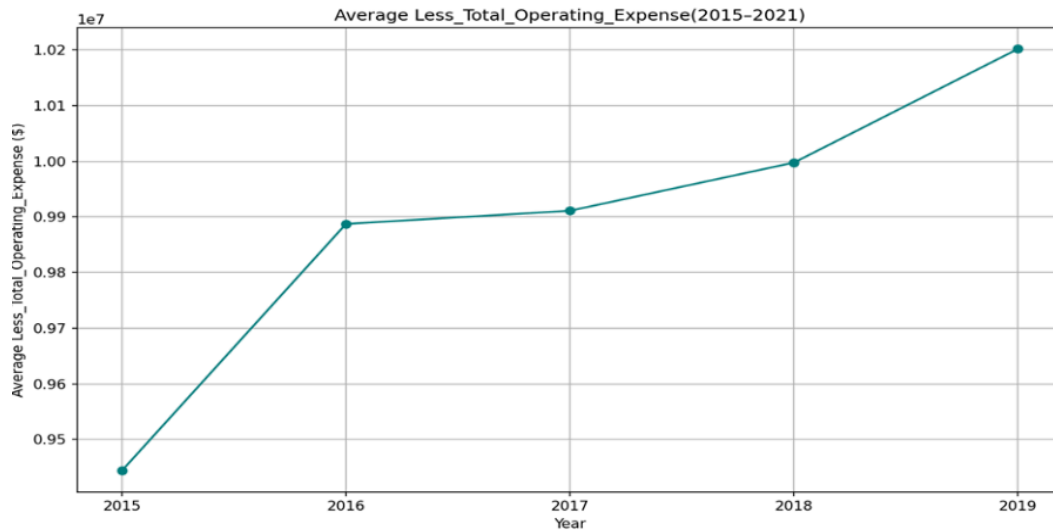
Revenue & Income

From 2015 to 2021, the financial performance of nursing homes reveals contrasting patterns across income, revenue, and expense. The average gross revenue rose steadily from 2015 to 2019, peaking around 2019, likely due to increased patient volumes. However, it declined in 2020 and 2021, potentially reflecting occupancy drops or disrupted services due to COVID-19. Operating expenses, on the other hand,

steadily increased through 2019, indicating rising costs in labor, supplies, and compliance. This trend of rising costs and plateauing revenues could drain profit margins and sustainability, particularly in facilities lacking financial reserves.

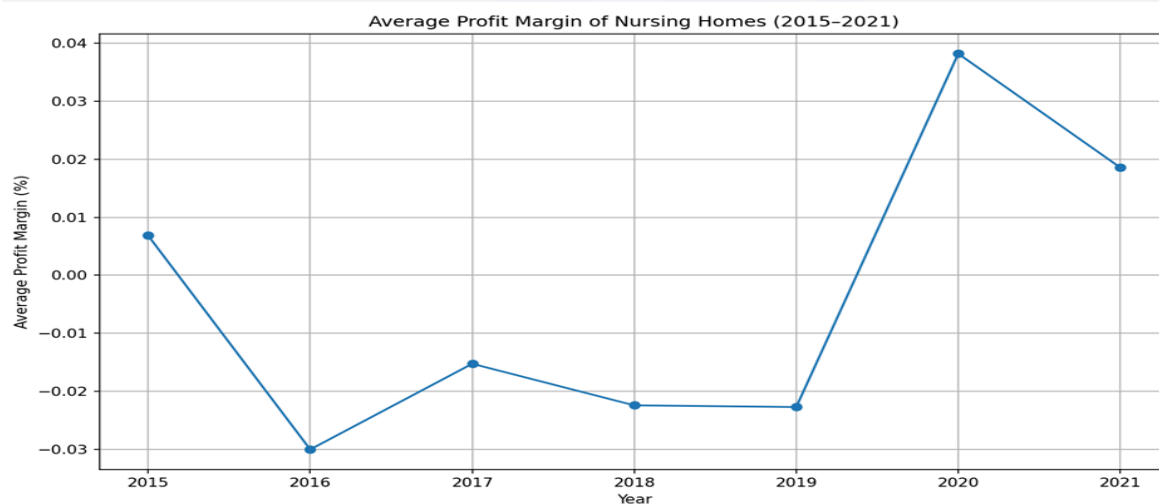
Net income, a key measure of profitability, shows a different story. It declined sharply from 2015 through 2018, hitting a low in 2018 before rebounding modestly in 2019. A dramatic spike occurred in 2020, which could be attributed to temporary COVID-related relief funding, hazard reimbursements, or deferred costs. However, this spike dropped again in 2021, suggesting that the income boost was not lasting. Overall, while nursing homes managed to grow revenue and absorb rising expenses pre-pandemic, the volatility in net income still shows vulnerability to the business model of SNF overall and shifting policy environments.





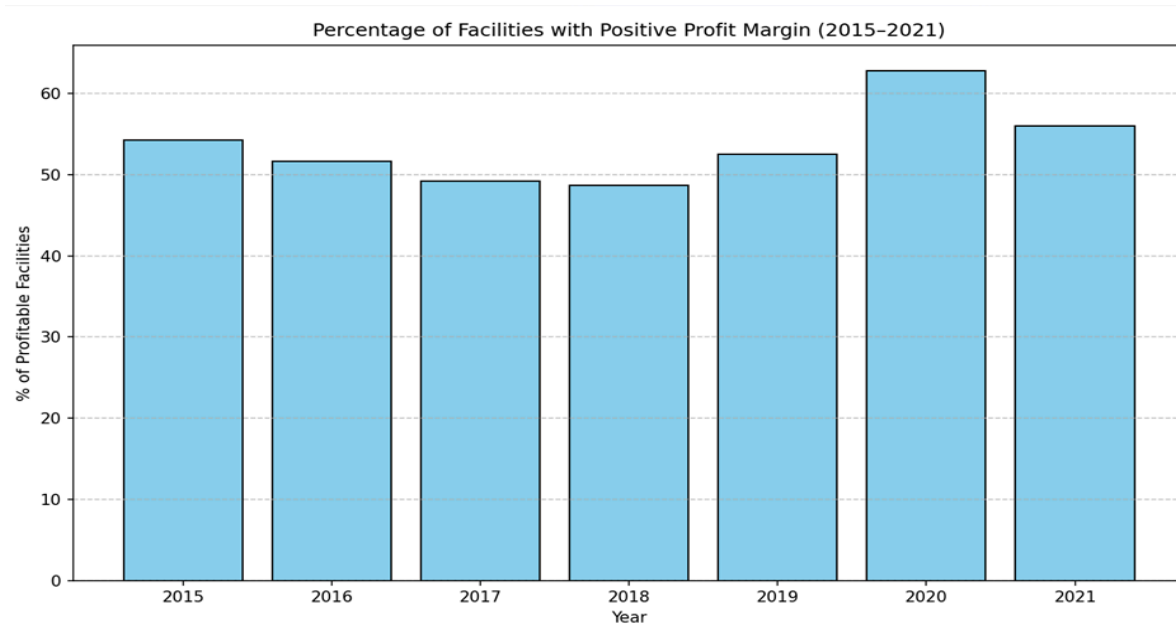
Profit Margin

Profit Margin is preferred to be used in this case to demonstrate correlated relationship between Gross Revenue and Net Income. It is calculated by performing a division of Net Income by Gross Revenue. As a result, data shows that it turned negative by 2016 and remained in the red through 2019. This means that, on average, facilities were spending more than they earned in those years.



In 2020, this trend reversed dramatically. Despite a decline in Gross Revenue due to lower occupancy and service disruptions during the COVID-19 pandemic, Net Income spiked significantly. This might be largely due to 'Total Other Income' received by the nursing homes during COVID-19 and we are assuming this income is from the federal relief funds provided through various government programs. These funds offset revenue

losses and reduced net operating costs, leading to a temporary improvement in Profit Margin, which peaked at nearly 4% in 2020. However, this improvement was not sustained, as both Net Income and Profit Margin declined again in 2021 as government aid was scaled back.



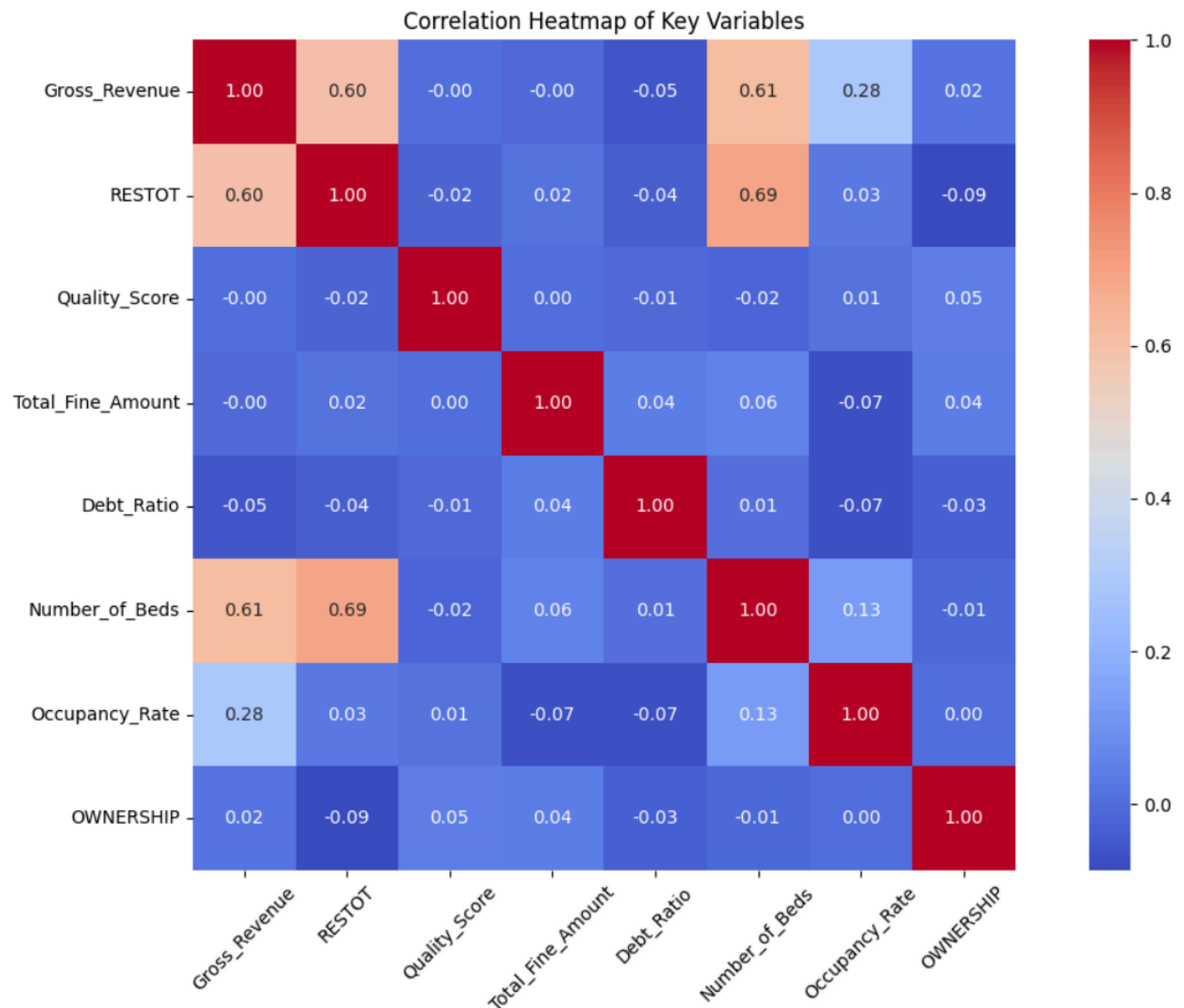
The percentage of nursing home facilities with positive profit margins fluctuated moderately between 2015 and 2021, ranging from 49% to 62%. Almost 50% of nursing homes every year were in profit. So we can say that it is a moderately profitable business.

2. Identify the influential factors affecting the financial performance of nursing homes.

We worked with a large dataset containing financial, operational, and regulatory variables from nursing homes. To find the most important features, we used a correlation heatmap and tested different variable combinations based on their R^2 scores. This helped us choose the final set of features: Number of Beds, BEDCERT (Certified Beds), Occupancy Rate, OWNERSHIP, Quality Score, Total Fine Amount, RESTOT (Total Patient Days), and Debt Ratio. We chose **Gross Revenue** as our target variable because it reflects the total income a facility generates. It is more stable than net income, especially during events like the COVID-19 pandemic, and better shows how operational and structural factors impact revenue.

In order to select columns that yield the highest impact on financial information, we performed correlation analysis which can be displayed by heatmap. After eliminating other attributes which are Cost Report contains interesting financial numbers such as

Gross Revenue, and other potential dependent variables (i.e. Number of Beds, Total Number of Rest Day)



Gross Revenue: Gross Revenue is the total income a facility earns before expenses or deductions. It represents the overall scale of services provided and is a key indicator of financial performance.

The final variables we used to predict Gross Revenue include:

- **Ownership:** This refers to whether a facility is for-profit, nonprofit, or government-run. Ownership type can influence management efficiency, resource allocation, and financial priorities, which in turn affect revenue.
- **Occupancy_Rate:** This measures how fully the facility uses its bed capacity. Higher occupancy typically leads to more revenue. It is calculated as:

$$\text{Occupancy Rate} = (\text{Total Patient Days} / (\text{Certified Beds} \times 365)) \times 100$$

- **BEDCERT (Certified Beds):** This indicates the total number of certified beds in a facility. It serves as a proxy for the facility's capacity and potential to generate revenue through patient services.
- **Number of Beds:** The physical number of beds available at the end of the reporting period. In some cases, this can differ from certified beds and may reflect underutilized capacity.
- **RESTOT (Total Patient Days):** This captures the total number of days all patients stayed in the facility during a year. It directly reflects service volume and is strongly associated with revenue.
- **Total_Fine_Amount:** This is the total amount of penalties a facility received in a given year. High fines may indicate compliance issues and can negatively impact financial outcomes.
- **Quality_Score:** This reflects the care quality at a facility based on inspections and outcome metrics. While higher quality may attract more residents, its impact on revenue can vary.
- **Debt_Ratio:** This indicates financial leverage and is calculated as:
Debt Ratio = Total Liabilities / Total Assets
 A higher ratio suggests more financial risk, which may limit resources for revenue-generating operations.

Multiple Linear Regression

We first applied a Multiple Linear Regression (MLR) model to examine the linear relationship between various features and Gross Revenue. This model helped us understand how each individual factor contributes to changes in facility revenue. The MLR model achieved an R^2 score of 0.558, meaning it was able to explain about 55.8% of the variation in gross revenue. While the model provided a moderate fit, it gave us clear coefficients for each feature, helping us interpret the direction and strength of their influence. Features like BEDCERT, Occupancy Rate, and RESTOT had a strong positive impact, while variables such as Debt Ratio had negative effects on gross revenue.

Linear Regression Results

R^2 Score: 0.5578

Coefficients

Intercept: -1150000

OWNERSHIP: 84709.37

RESTOT: 35484.17

BEDCERT: 119876.85

Total_Fine_Amount: -3.29

Quality_Score: 59609.88

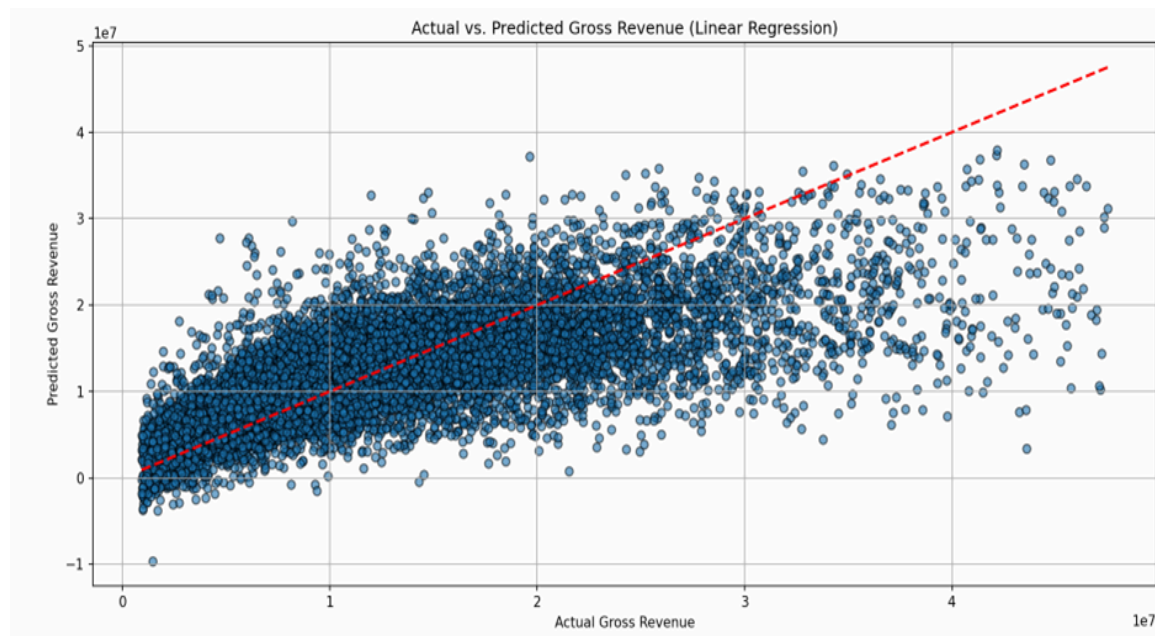
Debt_Ratio: -82894.76

Number_of_Beds: -36082.72

Occupancy_Rate: 110292.67

The linear regression model produced an R^2 score of 0.557, meaning approximately 55.7% of the variation in gross revenue can be explained by the selected independent variables. The model intercept was $-\$11.5$ million, indicating the baseline revenue when all other variables are zero. Key positive contributors included BEDCERT ($+\$119,800$), Occupancy Rate ($+\$110,292$), and OWNERSHIP ($+\$84,709$). Negative effects were observed from Debt Ratio ($-\$79,910$) and Total Fine Amount ($-\$3.18$).

We used OLS regression to examine coefficient significance. Most predictors had p-values < 0.05 , confirming statistical relevance. However, Quality Score, despite a coefficient of $+\$54,160$, showed a higher p-value, suggesting limited predictive value in this model. Overall, the results indicate that operational capacity and efficiency metrics play a critical role in influencing hospital gross revenue.



This scatter plot compares actual vs. predicted gross revenue using the Linear Regression model. Each point represents a facility. The red dashed line shows a perfect prediction. While most points follow the general trend, there's noticeable spread—especially at higher revenue levels—indicating that the model underpredicts high-revenue facilities and overpredicts some low ones. This highlights the limitations of linear regression in capturing complex patterns.

Random Forest Regressor

We then applied a Random Forest Regressor to capture more complex, nonlinear relationships between the features and Gross Revenue. This model showed improved performance over MLR, achieving an R^2 score of 0.6664, meaning it explained about 66.6% of the variation in gross revenue. Unlike linear regression, Random Forest provides feature importance scores instead of coefficients, helping us understand which variables had the greatest impact on the model's predictions. The most influential features were Number of Beds, Occupancy Rate, BEDCERT, and RESTOT, indicating that facility size and patient volume play a major role in determining gross revenue.

Random Forest Regression Results

R^2 Score: 0.6664

Feature Importances

OWNERSHIP: 0.0263

RESTOT: 0.1123

BEDCERT: 0.1375

Total_Fine_Amount: 0.0350

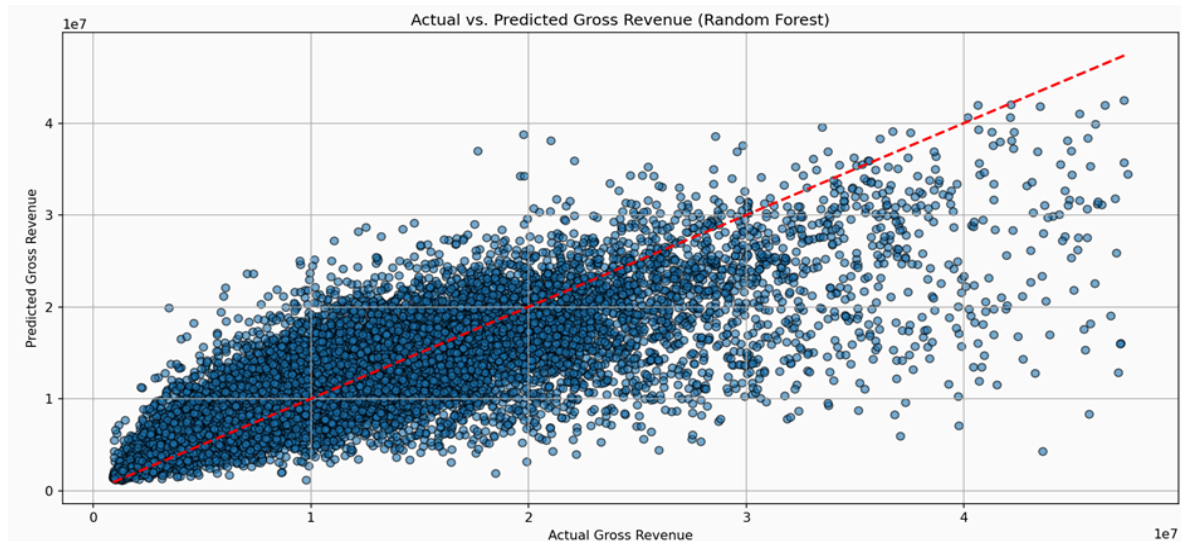
Quality_Score: 0.0132

Debt_Ratio: 0.0997

Number_of_Beds: 0.3190

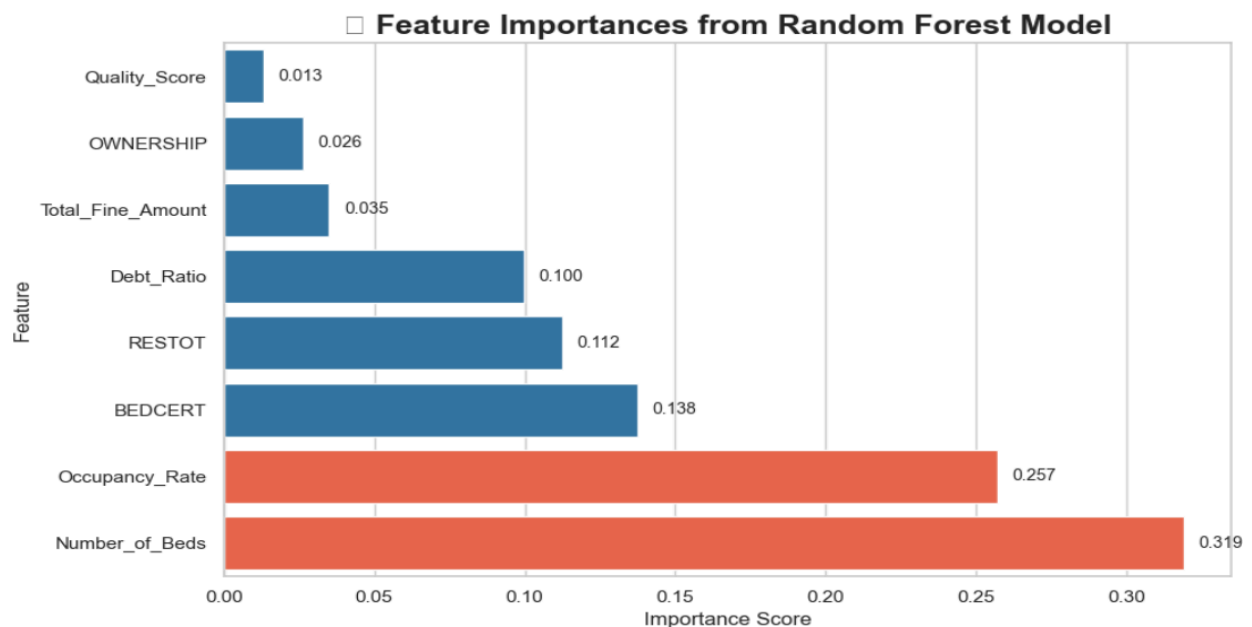
Occupancy_Rate: 0.2570

The Random Forest Regressor achieved a higher R^2 score of 0.6664, capturing more complex and nonlinear relationships between variables and gross revenue. Instead of coefficients, this model provides feature importance scores, revealing the most influential variables. Number of Beds was the top predictor, accounting for approximately 31.9% of the model's decision-making. This was followed by Occupancy Rate (25.7%), BEDCERT (13.8%), and RESTOT (11.2%). Features like Quality Score, Total Fine Amount, and OWNERSHIP had relatively lower importance. These results suggest that patient volume and facility size are the strongest drivers of revenue, while regulatory or quality-related variables have a more limited impact in this non-linear model.



This scatter plot shows actual vs. predicted gross revenue using the Random Forest model. Each dot is a facility, and the red dashed line indicates perfect prediction. Compared to linear regression, the points here are more tightly clustered around the line, especially in the low to mid revenue range, showing improved accuracy. However, the model still shows some spread at the higher end, though less than the linear model. This indicates better performance in capturing nonlinear patterns in the data.

3. Determine the most significant factors impacting the performance of nursing homes.



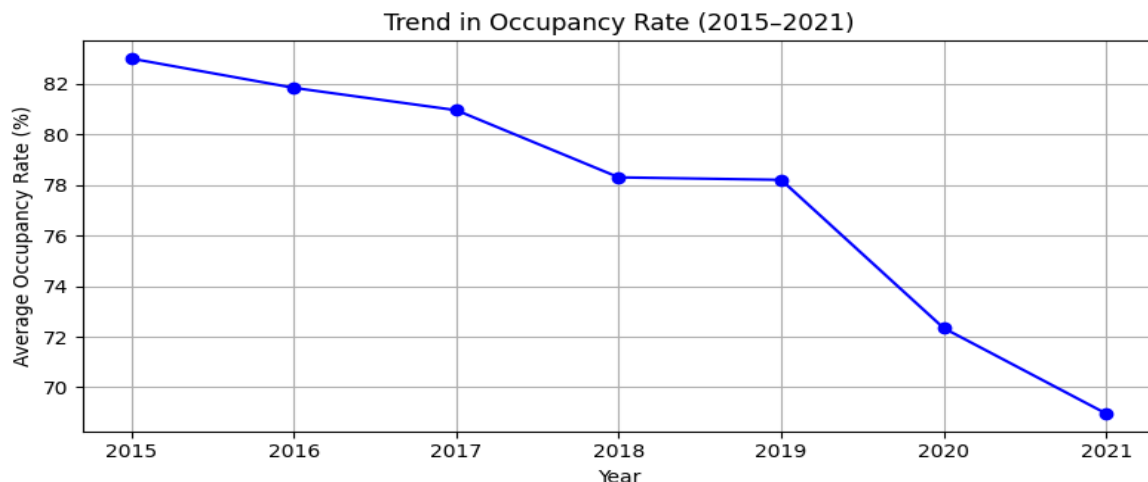
Taking a deeper look at statistical performance of each feature from Random Forest Regression, we selected most impacting features based on their importance. Based on the feature importance chart from the Random Forest model, the two most significant factors impacting the performance of nursing homes are:

Number_of_Beds (Importance: 0.319)

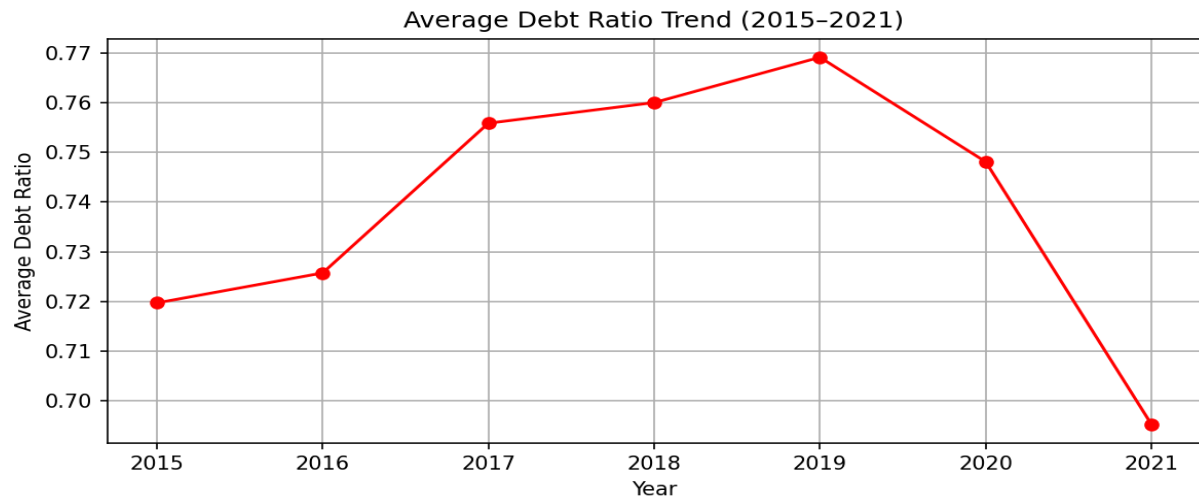
Occupancy_Rate (Importance: 0.257)

These two features were chosen because they have the highest importance scores among all variables in the model, indicating that they contribute the most to the predictive accuracy and decision-making of the Random Forest. Number_of_Beds reflects the capacity and scale of a nursing home. Larger facilities may have more resources, staff, and infrastructure, which can directly impact performance metrics. Occupancy_Rate measures how efficiently a nursing home is utilizing its capacity. High occupancy often correlates with operational effectiveness and financial stability, both of which are essential indicators of performance.

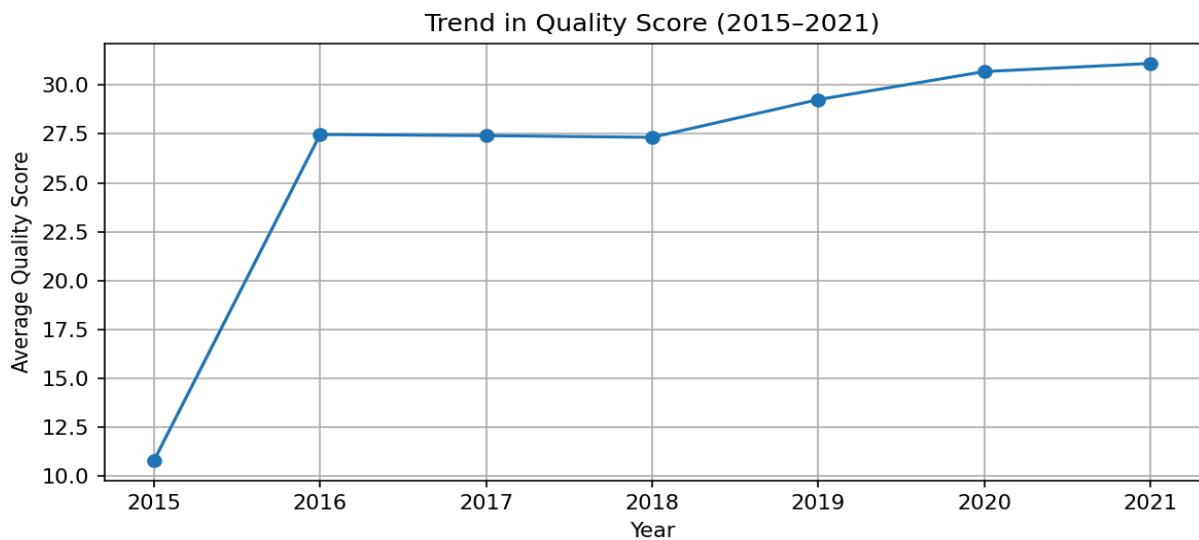
4. Describe the trends in performance and these influential factors.



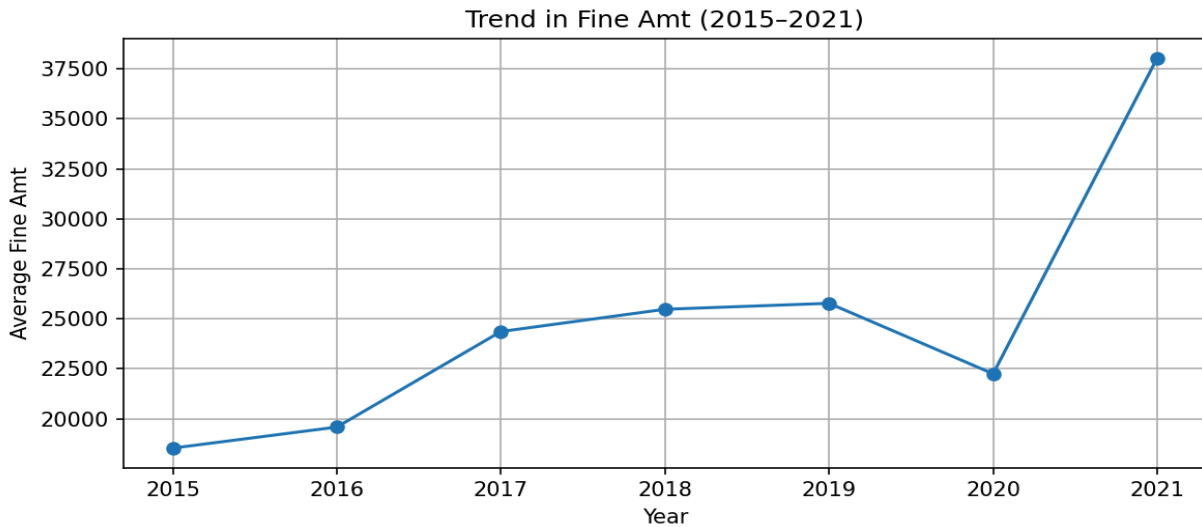
The number of people living in nursing homes decreased from 2015 to 2021. This decline was especially sharp in 2020-2021 due to the COVID-19 pandemic, staffing issues, and admissions being halted. Fewer families wanted to use nursing homes, and the homes themselves struggled to operate.



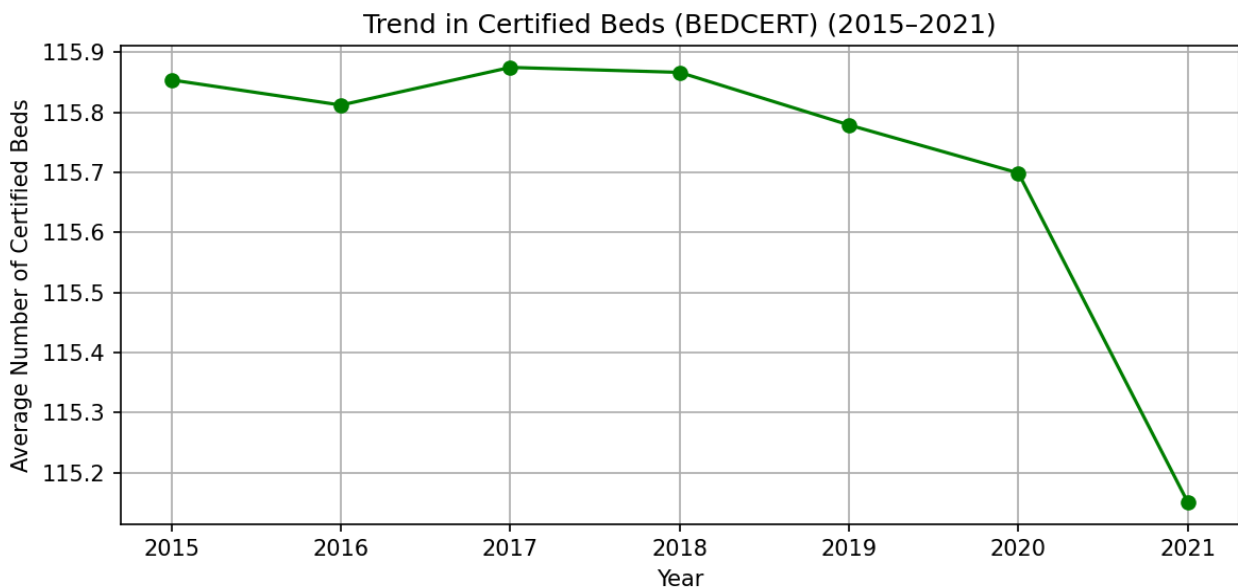
The company's debt levels went up from 2015 to 2019, meaning they were taking on more debt or having trouble making ends meet. But after 2019, they seemed to get their debt under control, possibly by paying off loans, getting pandemic-related financial help, or being more careful with their finances.



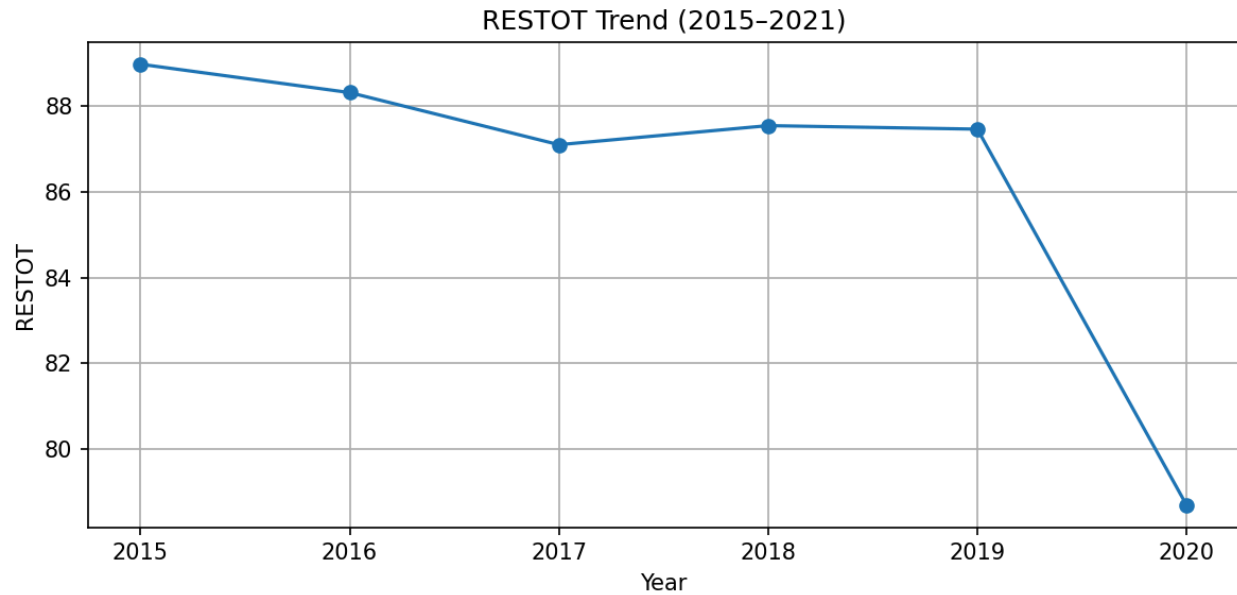
Quality scores went up from around 10 in 2015 to over 30 by 2021. This could mean care quality got worse, since higher scores might indicate poorer outcomes. On the other hand, the increase might be because the scoring system changed to include more metrics, making it hard to compare scores directly from year to year.



Penalty amounts steadily rose over time, with a sharp increase in 2021 likely caused by renewed inspections, tougher enforcement, and sanctions related to pandemic-era mismanagement.



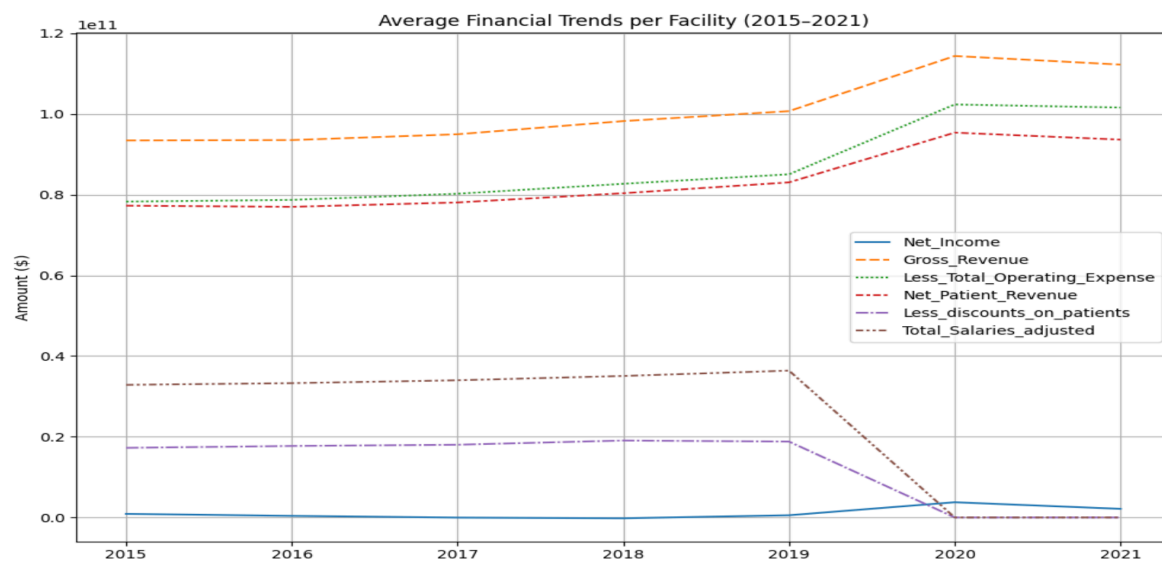
From 2015 to 2019, the number of certified beds per nursing home stayed pretty steady, showing that facilities had a consistent capacity. But in 2020, there was a noticeable drop, and in 2021, it plummeted. This might be because of COVID-19's impact, like some places closing or reducing their size, or maybe there's an issue with how the 2021 data was reported.



From 2015 to 2019, RESTOT showed a slight but steady decline, indicating gradual decreases in resident days. In 2020, there was a sharp drop, likely driven by the COVID-19 pandemic's impact on nursing home admissions and occupancy.

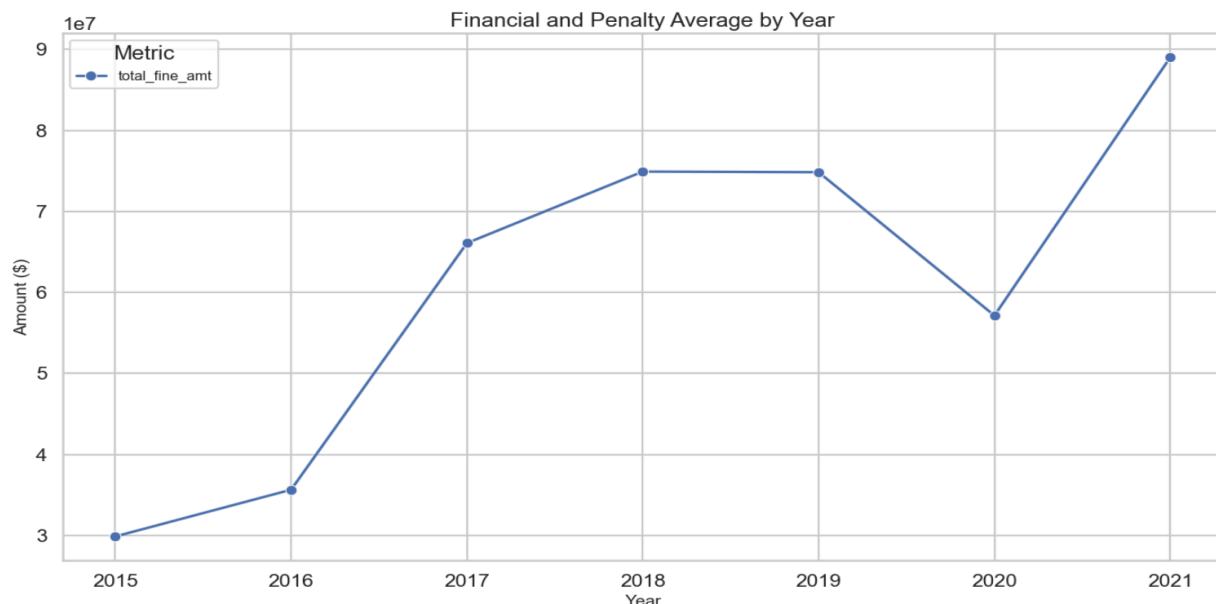
5. Analyze the impact of COVID-19.

The graph reveals significant shifts in hospital financials during the 2015–2021 period, with the most dramatic changes occurring in 2020. Both revenues and expenses rose sharply during the pandemic, likely driven by increased patient volume, emergency services, and pandemic-related costs. However, not all financial indicators followed the same trend.



Total Salaries (Adjusted) and Patient Discounts dropped significantly in 2020 and 2021, suggesting potential cost-cutting measures such as workforce reductions, salary cuts, or the suspension of non-essential services. The spike in Net Income during 2020 stands out when comparing changes across previous years. This anomaly may reflect temporary financial relief from government subsidies or programs which are designed to support healthcare providers during the COVID-19 crisis. By 2021, Net Income declined again, possibly indicating the reduction of such emergency funding. Additionally, the data show that Operating Expenses and Total Costs consistently moved together, highlighting that operational spending remains a major component of total expenditures. The rise in these expenses in 2020 may reflect the increased burden of acquiring protective gears for COVID-19, maintaining staffing levels, and implementing safety protocols.

- **Impact on Fine and Penalty**

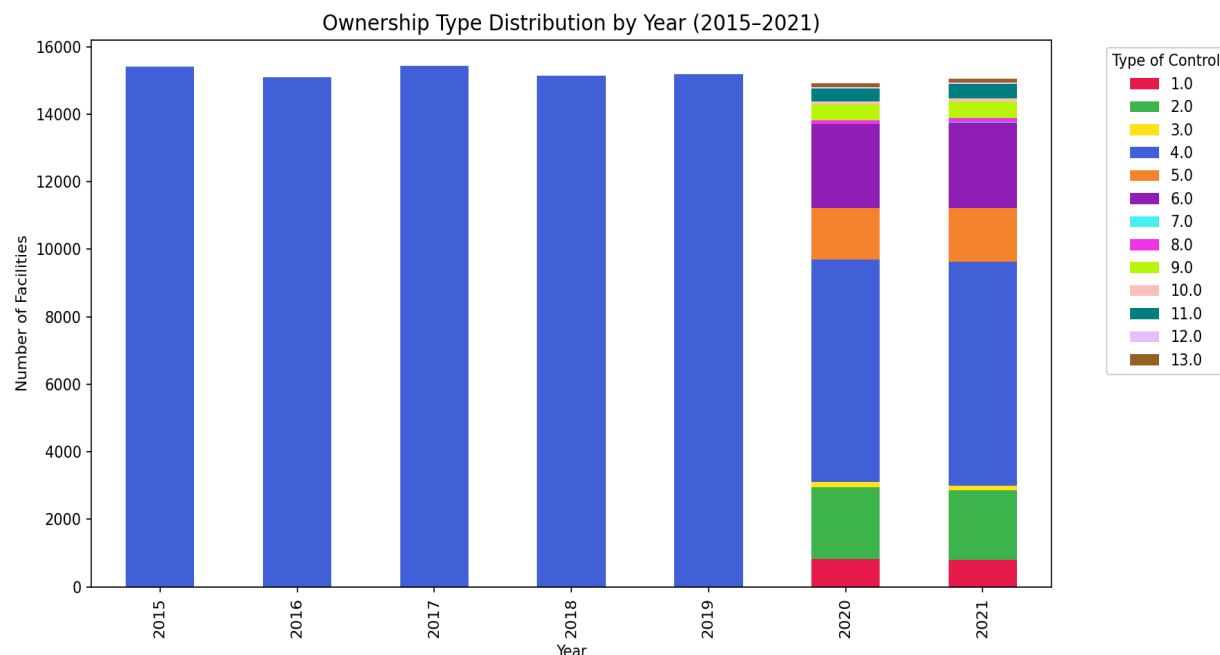


In 2020, there was a clear increase in fines and penalties given to healthcare providers, which lines up with the outbreak of the COVID-19 pandemic. This rise likely came from a mix of stricter regulations, rapid changes in how hospitals operated, and the overwhelming pressure the healthcare system faced. Hospitals had to quickly adjust to new safety guidelines, deal with staff shortages, and handle financial uncertainty. On top of that, federal support may have led to more audits that exposed more violations. It is important to understand that this spike does not necessarily mean healthcare providers were being negligent, it may just reflect the stress and confusion of the time. When looking at this data, financial institutions should view the 2020 spike as a temporary

effect of the pandemic, not a sign of long-term issues, especially if future observation shows a downtrend in penalty.

- **Impact of Ownership Type**

The stark shift in ownership type distribution of nursing homes in 2020 and 2021 likely reflects major structural and regulatory changes triggered by the COVID-19 pandemic. Prior to 2020, most facilities were classified under Type 4 (Proprietary-Corporation), but the sudden diversification suggests a combination of improved reporting standards, ownership transfers, and government intervention. Increased scrutiny of for-profit chains during the pandemic led to heightened transparency requirements and possibly reclassification of ownership types that were previously lumped under Type 4. Additionally, financial stress, staffing shortages, and legal pressures may have caused many corporate facilities to close or be sold to government or nonprofit entities. These changes, along with updates to CMS data reporting, likely account for the broader range of ownership categories seen in the later years of the dataset.



Discussion:

Our analysis shows that nursing home revenue is mostly affected by how well the facility is run and how many residents it serves. Important factors in both models were the Number of Beds, Occupancy Rate, Certified Beds (BEDCERT), and Total Patient Days (RESTOT). These show how big and busy a facility is, which helps it earn more money. Ownership type had some effect in the linear model, but it didn't matter much in the

Random Forest model—this means the way a facility is managed is more important than who owns it. Debt Ratio had a negative effect, meaning too much debt can hurt a facility's income. In 2020, revenue went up because of government help during COVID-19, but this was temporary and dropped back to normal in 2021.

Between the two models, Random Forest predicted better and handled complex relationships well. On the other hand, linear regression was easier to understand and helped explain how each factor affected revenue. Instead of picking just one model, using both is a good idea—Random Forest for better predictions and Linear Regression for easier understanding. Together, they give a clearer picture for making good decisions in the nursing home industry.

Recommendations

Based on our findings, we suggest the following actions for different groups who are involved with or affected by nursing home operations:

1. For Investors:

Investors should look for nursing homes that are for-profit, have a high occupancy rate, and carry low levels of debt. These homes are more likely to be financially stable and profitable. Facilities with many fines or declining bed counts should be avoided, as these are signs of operational problems or possible closure. Investors should also be cautious about relying too heavily on one-time pandemic-related financial boosts when evaluating a facility's long-term potential.

2. For Nursing Home Operators:

Operators should focus on improving day-to-day efficiency, keeping beds filled, and carefully managing their finances. Reducing unnecessary expenses and staying on top of debt can help maintain healthy profit margins. It is also important to train staff well and follow all regulations closely to avoid costly penalties. Improving quality of care not only helps residents but can also lead to better financial results over time.

3. For Policymakers and Regulators:

Government agencies should ensure that financial assistance goes to the nursing homes that need it most—especially those with high occupancy but limited financial resources. More accurate and transparent reporting of ownership types is also necessary so that policy decisions are based on reliable data. In future emergencies like pandemics, flexible policies that allow facilities to respond quickly and still meet safety standards will be essential.

Conclusion:

This project explored financial performance in U.S. nursing homes from 2015 to 2021 using data-driven models. We found that gross revenue is most affected by how well facilities manage capacity and patient volume—especially factors like bed count, occupancy, and total days of care. Costs continued to rise, and while revenue grew pre-pandemic, the industry faced instability during COVID-19. Relief funds temporarily boosted income in 2020, but this gain did not last.

Our models identified operational efficiency and scale as the strongest predictors of revenue, while ownership type and penalties had a smaller influence. Facilities that optimize their available resources, manage debt, and maintain high occupancy are more likely to stay financially sustainable. As the need for long-term care grows, improving these core operational areas will be essential for the future of the nursing home industry.

Overall, the nursing home industry remains essential but fragile. Stronger operations, better data reporting, smarter investment choices, and targeted policy support are all needed to keep these facilities running effectively. As the population continues to age, ensuring the financial health and care quality of nursing homes should remain a top priority for everyone involved.

Appendices:

Group and average before merging

```
quality_avg = quality_df.groupby(['provnum',  
'Year'])['Quality_Score'].mean().reset_index()
```

=== Group and sum fines per facility/year

```
penalties_df['fine_amt'] = pd.to_numeric(penalties_df['fine_amt'], errors='coerce')  
penalty_sum = penalties_df.groupby(['provnum', 'Year'])['fine_amt'].sum().reset_index()  
penalty_sum.rename(columns={'fine_amt': 'Total_Fine_Amount'}, inplace=True)
```

=== Merge datasets

```
df = cost_df.merge(provider_df[['provnum', 'OWNERSHIP',  
'BEDCERT', 'RESTOT', 'Year']], on=['provnum', 'Year'], how='left')  
df = df.merge(penalty_sum, on=['provnum', 'Year'], how='left')  
df = df.merge(quality_avg, on=['provnum', 'Year'], how='left')
```

=== Ensure numeric columns

```
numeric_cols = ['Net_Income', 'Gross_Revenue',  
'RESTOT','Number_of_Beds','Less_Total_Operating_Expense','Quality_Score',  
'Total_liabilities','Total_Assets','BEDCERT','Total_Days_Total', 'Total_Fine_Amount']  
df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric, errors='coerce')
```

=== Calculate Debt_Ratio

```
df['Debt_Ratio'] = df['Total_liabilities'] / df['Total_Assets']
```

=== Impute Total_Fine_Amount with median

```
median_fine = df['Total_Fine_Amount'].median()  
df['Total_Fine_Amount'] = df['Total_Fine_Amount'].fillna(median_fine)
```

=== Drop rows missing Net_Income or Gross_Revenue

```
df = df.dropna(subset=['Net_Income', 'Gross_Revenue'])
```

=== Encode ownership

```
df['OWNERSHIP'] = df['OWNERSHIP'].fillna('Unknown')  
df['OWNERSHIP'] = LabelEncoder().fit_transform(df['OWNERSHIP'])
```

=== Impute remaining features

```
imputer = SimpleImputer(strategy='median')  
df[['Net_Income', 'Gross_Revenue','Less_Total_Operating_Expense','RESTOT',  
'Number_of_Beds','BEDCERT','Quality_Score', 'Debt_Ratio','Total_Days_Total']] =  
imputer.fit_transform(  
    df[['Net_Income', 'Gross_Revenue','Less_Total_Operating_Expense','RESTOT',  
'Number_of_Beds','BEDCERT','Quality_Score','Debt_Ratio','Total_Days_Total']]  
)
```

=== Derived features

```
df['Profit_Margin'] = (df['Net_Income'] / df['Gross_Revenue']) * 100  
df['Occupancy_Rate'] = np.where(  
    df['BEDCERT'] > 0,  
    (df['Total_Days_Total'] / (df['BEDCERT'] * 365)) * 100,  
    np.nan  
)
```

=== Impute Occupancy_Rate


```
df['Occupancy_Rate'] = df['Occupancy_Rate'].fillna(df['Occupancy_Rate'].median())
```

```
=====
```

=== Train/test split

```
X = df[features]
```

```
y = df[target]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

=== Linear Regression

```
lr_model = LinearRegression()
```

```
lr_model.fit(X_train, y_train)
```

```
y_pred_lr = lr_model.predict(X_test)
```

```
print("\n Linear Regression Results")
```

```
print("R2 Score:", round(r2_score(y_test, y_pred_lr), 4))
```

```
print("RMSE:", round(np.sqrt(mean_squared_error(y_test, y_pred_lr)), 2))
```

```
print("\n Coefficients:")
```

```
for name, coef in zip(X.columns, lr_model.coef_):
```

```
    print(f"{name}: {coef:.4f}")
```

```
X_ols = sm.add_constant(X) # add intercept term
```

```
ols_model = sm.OLS(y, X_ols).fit()
```

```
print("\nOLS Regression Summary:")
```

```
print(ols_model.summary())
```

```
=====
```

=== Train/test split

```
X = df[features]
```

```
y = df[target]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

=== Random Forest Regression

```
rf_model = RandomForestRegressor(n_estimators=100, max_depth=None,  
random_state=42)
```

```
rf_model.fit(X_train, y_train)
```

```
y_pred_rf = rf_model.predict(X_test)
```

```
print("\nRandom Forest Regression Results")
print("R² Score:", round(r2_score(y_test, y_pred_rf), 4))
print("RMSE:", round(np.sqrt(mean_squared_error(y_test, y_pred_rf)), 2))
```

```
print("\nFeature Importances:")
for name, importance in zip(X.columns, rf_model.feature_importances_):
    print(f'{name}: {importance:.4f}')
```

=====

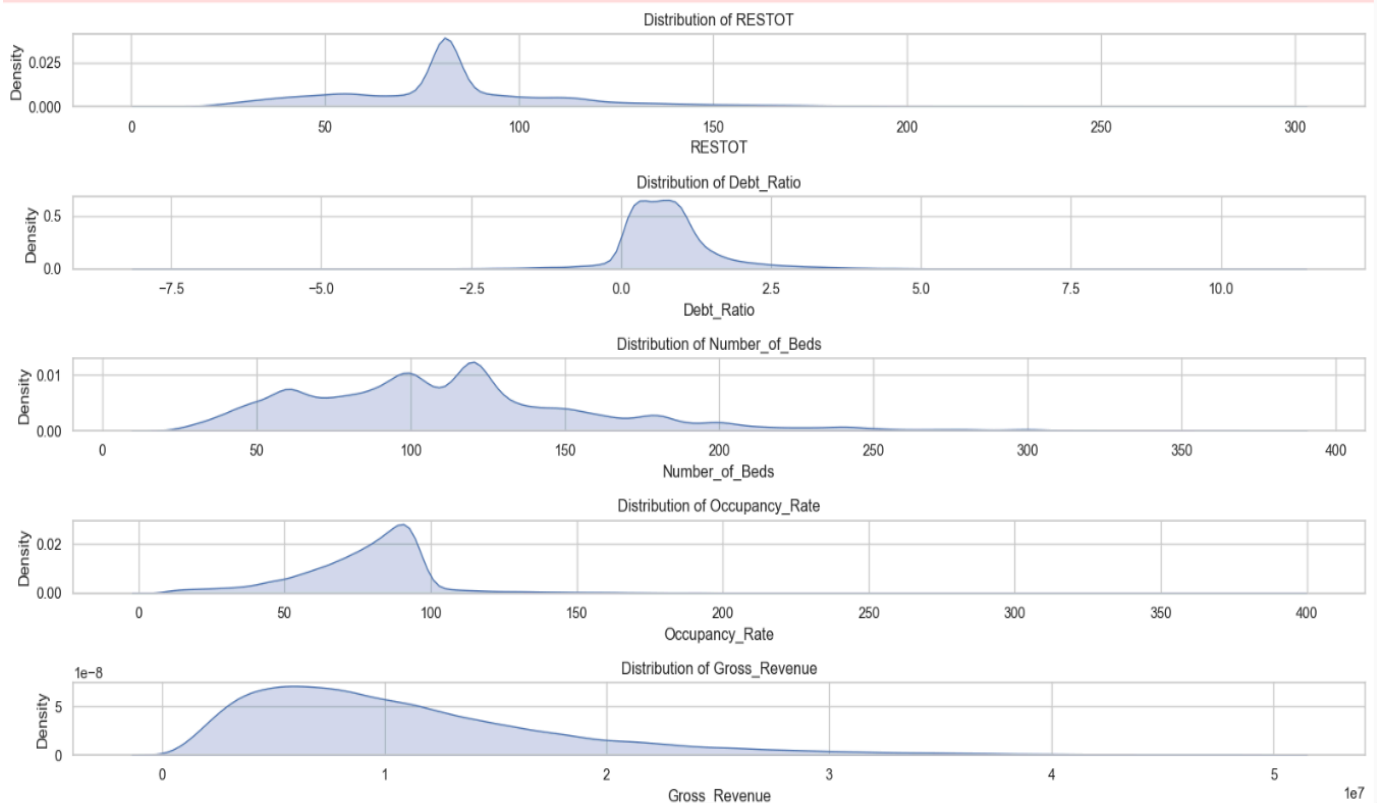
Distribution Overview Code and Output

```
for i, col in enumerate(df_subset.columns):
    plt.subplot(num_cols, 1, i + 1)
    sns.kdeplot(df_subset[col].dropna(), shade=True)
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
```

```
plt.tight_layout()
plt.show()
```

shade is now deprecated in favor of fill, setting fill=True.
This will become an error in seaborn v0.14.0; please update your code.

```
sns.kdeplot(df_subset[col].dropna(), shade=True)
```



=====

Define columns to analyze for correlation heatmap

```
corr_columns = ['Net_Income', 'Gross_Revenue', 'Year', 'provnum',
                'PROVNAME', 'OWNERSHIP', 'BEDCERT', 'RESTOT']
```

Loop through each year's merged dataframe

```
for year, df in merged_dataframes.items():
    print(f"Generating correlation heatmap for {year}...")

    # Ensure required columns are present
    if all(col in df.columns for col in corr_columns):
        # Drop rows with missing values in the correlation columns
        subset = df[corr_columns].dropna()

        # Only compute correlation if there's enough data
        if len(subset) > 10:
            # Compute correlation matrix
            corr = subset.corr()

            # Plot heatmap
            plt.figure(figsize=(8, 6))
            sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f", square=True)
            plt.title(f'Correlation: Ratings vs Financials ({year})')
            plt.tight_layout()
            plt.show()
        else:
            print(f"Not enough data to compute correlation for {year}.")
    else:
        print(f"One or more columns missing in {year} dataset.")
```

References:

- Scikit-learn library for machine learning
- Statsmodels for regression analysis
- Seaborn and Matplotlib for graphs
- U.S. Department of Health and Human Services reports on COVID-19 funding